Visual abstract: from heterogeneous evidence to a persistent component-level digital twin

One GPT-6 Astra/Codex workflow repeatedly observes, models, validates, revises, reuses, and composes task-level outputs.

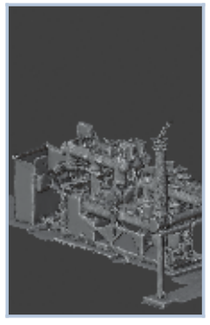
A · Evidence

Field photograph



visible component evidence

Coarse reference



registered photogrammetric geometry

B · Agentic loop

GPT-6 Astra + Codex

3D engineering agent

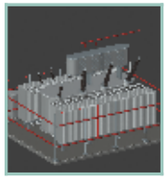
inspect evidence
write Blender/Python
validate · diagnose · revise

C · Heterogeneous reconstruction tasks

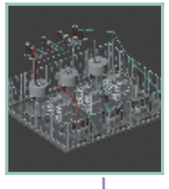
Arrester



Transformer



Capacitor bank

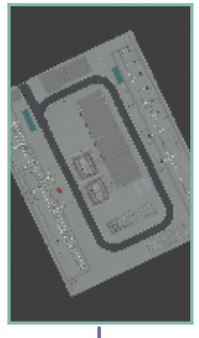


220 kV GIS



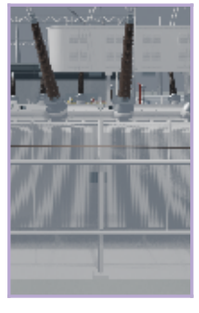
D · System-level composition

Integrated station



D38.2 core geometry

Inspection semantics



D41 transformer accessory layer

Persistent engineering state

MASTER / instances · deterministic validators · immutable batches · human-steerable revision

12 task families	38 core batches	37,153 scene objects	≈9.5× descriptive reuse	D41: 79 meshes / 17 classes × T1/T2

Figure 10 · Real project evidence and renders. The visual abstract summarizes the paper's core claim: task-level reconstructions remain part of one persistent, revisable engineering state.